Muhammad Muizz Zafar

muizzafar11@gmail.com · 905-805-9142 · linkedin.com/in/muizz-zafar/

Software Engineer | DevOps & QA Testing | Cloud Infrastructure | Full Stack Web Developer

Skills

Code Languages:	Java, Python, TypeScript, GoLang, SQL, JavaScript, Kotlin, C#, C++, HTML/CSS, PHP
Tools/Frameworks:	AWS CDK, CloudFormation, Docker, CloudWatch, Apache Spark, Flask, GraphQL, Git, Linux
Testing Expertise:	QA Automation, Test Case Design, Agile Testing, Regression Testing, Defect Tracking

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

Work experience

Software Developer — May 2024 - PRESENT
Elections Canada — Ottawa, ON

- Built automated **Python ETL testing pipelines** and Azure DevOps **CI/CD workflows** to verify production data movement across 20M+ records in multiple environments, reducing manual validation effort.
- Used **Apache Spark** to profile, transform, and validate datasets of up to **5+ GB**, accelerating bulk data checks and anomaly detection across staging and production environments.
- Built internal data analytics **dashboards** with **Python**, **Streamlit**, and **Plotly Dash** to surface pipeline health, operational metrics, and data quality issues for senior management.
- Coordinated version-controlled database deployments, improving traceability and reliability of **production data infrastructure**.
- Collaborated with cloud engineers and DevOps teams to improve **distributed data workflows**, release quality, and production reliability.
- Configured **Firewall hosts** and managed **IP** whitelisting to enable secure access for jumpbox access to UAT and PROD environments.

Full Stack Developer Intern — September 2023 - September 2024
University of Ottawa — Ottawa, ON

- Delivered production updates to the University of Ottawa emergency alert platform using **C#** and **Angular**, supporting a high-traffic service used by over **20,000** students daily.
- Designed and executed **unit, end-to-end tests** using **Jasmine**, **Cypress**, and **JUnit**, validating new frontend and backend features.
- Developed RESTful APIs in **.Net Core (C#)** for updating users information and adding new events that helped improve user satisfaction. Implemented CRUD APIs for user post system that helped increase app engagements and data traffic by **25%**
- Enhanced backend functionality for seamless API-to-frontend interactions in a microservices architecture.
- Collaborated with engineering teams in an **Agile sprint cycle** to review test results, ensure feature reliability, and fix UI/API integration bugs.

Cloud Developer Intern — May 2023 - August 2023
Trend Micro Inc. — Ottawa, ON

- Developed serverless monitoring workflows using **AWS Lambda, S3, and CloudWatch** to analyze **50,000+ artifacts per day** and support automated vulnerability scanning pipelines for high-volume cloud artifacts.
- Defined and maintained cloud infrastructure using **Golang**, **AWS CDK**, through **CloudFormation** to provision **50+ AWS resources** and improve repeatability of service deployments.
- Implemented concurrent processing logic in **GoLang** using goroutines to improve throughput and strengthen fault handling for **distributed cloud data processing** workloads.
- Integrated vulnerability scanning tools (**Anchore, Grype**) with AWS services to automate security checks and operational monitoring across **50,000+ cloud artifacts per day**.

Cloud Developer Intern — September 2022 - December 2022
Trend Micro Inc. — Ottawa, ON

- Built a serverless automation pipeline using **AWS Lambda, S3, and Athena** to process incoming cloud data events, query metadata, and trigger downstream updates, reducing manual operational effort.
- Configured **Splunk** and **AWS CloudWatch** monitoring, alarms, dashboards, and alerts to track **15+ services** and improve observability of production cloud infrastructure.
- Developed **Docker-based integration tests** and unit tests to improve reliability of distributed serverless data pipelines.

Full Stack Developer Intern — January 2022 - April 2022
Applied Mind Inc. — Ottawa, ON

- Built **Docker** containers and automated event ingestion into **PostgreSQL** using **Python** and **GraphQl**, reducing testing time and improving deployment readiness.